International Journal of Forecasting 3 (1987) 313–328
North-Holland

OPTIMAL ERROR AND GMDH PREDICTORS
A Comparison with Some Statistical Techniques

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Abstract: This paper presents some results obtained in time series forecasting using two nonstandard approaches and compares them with those obtained by usual statistical techniques. In particular, a new method based on recent results of the General Theory of Optimal Algorithms is considered. This method may be useful when no reliable statistical hypotheses can be made or when a limited number of observations is available. Moreover, a nonlinear modelling technique based on Group Method of Data Handling (GMDH) is also considered to derive forecasts. The well-known Wolf Sunspot Numbers and Annual Canadian Lynx Trappings series are analyzed; the Optimal Error Predictor is also applied to a recently published demographic series on Australian Births. The reported results show that the Optimal Error and GMDH predictors provide accurate one step ahead forecasts with respect to those obtained by some linear and nonlinear statistical models. Furthermore, the Optimal Error Predictor shows very good performances in multistep forecasting.

Keywords: Optimal algorithm, Time series, GMDH, Application, Comparative methods, Lead time.

1. Introduction

As is well known, time series forecasting has been almost always dealt with in a statistical framework, where the real time series is assumed to be a realization of a stochastic process of partially known statistics. Of course, the basic hypothesis underlying any statistical approach is that the length of the observation interval of the realization of the process is sufficient to ensure the reliability of statistical tests, which often only have good properties asymptotically. Actually, real time series of quite short length (less than one or two hundred data) are frequently met in the literature, especially in econometric and ecological contexts. In these cases it may happen that a statistical approach requires assumptions (such as stationarity, ergodicity, linearity, distribution structure) which cannot be easily and reliably tested, so that it may be more appropriate to use non statistical methods.

This paper presents and compares the forecasting results obtained by two alternative approaches (Optimal Error Algorithm and GMDH) with those of several statistical linear and nonlinear models. The Optimal Error Predictor [Milanese and Tempo (1985), Milanese et al. (1984), Belforte et al. (1982)] is based on the General Theory of Optimal Algorithms, which has been mainly developed in connection with problems of interpolation, approximation of function, nonlinear optimization [Traub and Wozniakowski (1980), Micchelli and Rivlin (1977)] and system parameter and state

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estimation [Milanese et al. (1984), Belforte et al. (1982)]. This approach does not require any statistical hypotheses and assumes that the noise process is unknown but bounded as the only a priori knowledge on the time series.

The GMDH, originally proposed in Ivakhnenko (1970) and later improved in Duffy and Franklin (1975), Ikeda et al. (1976), Tamura and Kondo (1980) and Gallo et al. (1985), consists essentially of a procedure which builds a polynomial nonlinear approximating model of the series giving rise to a Volterra expansion [Gabor et al. (1961)] in which only some terms have been selected according to some criterion. It is especially suited for cases when relatively little data are available and no a priori information is available on the structure of the mechanism generating the series.

In section 1 a brief account of the General Theory of Optimal Algorithms is given and its application to the problem of time series forecasting is introduced. In this context a time series may be considered as a 'problem element' belonging to a given space whose structure represents the a priori knowledge about the series. The goal is to forecast the time series by means of an optimal error predictor according to a minimax criterion using the available information.

Section 2 presents the GMDH procedure and focuses on the application of the original idea [Ivakhnenko (1970)] to the analysis of univariate time series. In particular, the PRESS criterion is used both for the determination of partial model structures and for the selection of intermediate variables. The multistage procedure leads to the structure and the parameters of the series model, which is successively used to forecast future values of the series.

In section 3 three time series are used for comparison of results: the Annual Canadian Lynx Trappings [Campbell and Walker (1977)], the Yearly Wolf Sunspot Numbers [Waldmeier (1961)] and the Australian Births [McDonald (1981)]. Several linear and nonlinear statistical models are compared with the Optimal Error and GMDH predictors on the basis of mean square and maximum prediction error criteria.

2. Optimal error predictors

The Theory of Optimal Algorithms provides a general framework for studying problems which cannot be solved exactly. The goal is to recover a given transformation S of an unknown problem element x, belonging to a fixed class, by an algorithm A using information y possibly corrupted by noise. The study of error contaminated information may be relevant in practical problems when the 'true information' $I(x)$ is unknown and only the 'approximate information' y is available, being the error between true and approximate information bounded [Milanese and Tempo (1985), Micchelli and Rivlin (1977)]. An 'optimal error algorithm' A_0 is the best algorithm in the sense that it minimizes the approximation error between $S(x)$ and $A_0(y)$ for all problem elements x and approximate information y.

In the context of time series forecasting the problem element x is a time function $x(\cdot)$ belonging to a given space X. Let K, a subset of X, represent the available a priori information on $x(\cdot)$. The problem is to forecast the future value of x at a fixed time $(m+1)$ when the information $I(x)$ available at time m is made of the observations of x; then the solution S and information I are

$$S(x) = \{x(m+1)\}, \tag{1}$$

$$I(x) = \{x(1), \ldots, x(m)\}. \tag{2}$$

The problem of determining an appropriate structure of the space X is the same as that of model structure selection in statistical identification contexts and it can be solved, for example, by using

prediction error criteria [Akaike (1969), Allen (1974), Parzen (1974), Kashyap and Rao (1976)]. A structure of the space X representing both autoregressive and deterministic (polynomial, harmonic,...) components is sufficiently general to represent a broad class of time series [Milanese et al. (1984)]. In fact, we will consider X as the space of discrete time functions $x(\cdot)$ defined on the interval $[q+1,\ m+1]$

$$X=\left\{x(\cdot): x(k)=\sum_{i=1}^{p} a_i d_i(k)+\sum_{i=1}^{q} b_i\delta(k-i)y(k-i)+\sum_{i=q+1}^{m+1} e(i)\delta(k-i);\right.$$

$$\left. k=q+1,\ldots,m+1\right\}, \tag{3}$$

where $d_i(\cdot)$ are deterministic discrete known functions, $y(k-i)$ are the observations of $x(\cdot)$, a_i and b_i are real unknown coefficients, $e(i)$ are coefficients of a discrete noise, p and q are fixed integers and $\delta(\cdot)$ is the unit impulse function. The observations of the time series $x(\cdot)$ at times $1,\ldots,m$ are

$$y=\{y(1),\ldots,y(m)\}, \tag{4}$$

where $y(1),\ldots,y(q)$ are the initial conditions of the series $x(\cdot)$ defined in (3). It should be remarked that in the context of time series forecasting the observations y are usually not corrupted by noise; thus it follows that $I(x)$ is exactly known

$$I(x)=y. \tag{5}$$

The problem of forecasting $x(\cdot)$ at time $m+1$ can be expressed as

$$\hat{x}(m+1)=A(y), \tag{6}$$

where $\hat{x}(m+1)$ denotes the prediction of $x(m+1)$. Suppose that the discrete noise samples $e(i)$ are unknown but bounded

$$|g(i)e(i)|\leqslant\xi, \qquad i=q+1,\ldots,m+1, \tag{7}$$

where $g(i)$ are fixed positive weights and ξ is a positive constant. The structure of the subset K of the space X, which depends on the a priori knowledge on the specific series, is defined as

$$K=\{x(\cdot)\in X: |g(i)e(i)|\leqslant\xi;\ i=q+1,\ldots,m+1\}. \tag{8}$$

An Optimal Error predictor A_0 is defined as an algorithm which minimizes the maximum possible prediction error over all possible $x\in K$ having the same information $I(x)$

$$\max_{x\in K}|x(m+1)-A_0(I(x))|=\min_{A}\max_{x\in K}|x(m+1)-A(I(x))|. \tag{9}$$

It has been shown that such an Optimal Error predictor is given by [Milanese and Tempo (1985), Milanese et al. (1984)]

$$A_0(y) = \hat{x}(m+1) = \tfrac{1}{2}\left\{ \max_{x \in T(y)} x(m+1) + \min_{x \in T(y)} x(m+1) \right\}, \qquad \text{where} \tag{10}$$

$$T(y) = \{ x(\cdot) \in K : x(i) = y(i);\ i = q+1, \ldots, m \}. \tag{11}$$

Using (3) and (11) the Optimal Error predictor (10) becomes

$$\hat{x}(m+1) = \tfrac{1}{2} \max\left\{ \sum_{i=1}^{p} a_i d_i(m+1) + \sum_{i=1}^{q} b_i y(m+1-i) \right\} +$$

$$\left| y(k) - \sum_{i=1}^{p} a_i d_i(k) - \sum_{i=1}^{q} b_i y(k-i) \right| \leqslant \xi / g(k), \qquad k = q+1, \ldots, m$$

$$+ \tfrac{1}{2} \min\left\{ \sum_{i=1}^{p} a_i d_i(m+1) + \sum_{i=1}^{q} b_i y(m+1-i) \right\}.$$

$$\left| y(k) - \sum_{i=1}^{p} a_i d_i(k) - \sum_{i=1}^{q} b_i y(k-i) \right| \leqslant \xi / g(k), \qquad k = q+1, \ldots, m \tag{12}$$

As relation (12) is composed of a maximization and a minimization of a linear functional over a set described by linear inequalities, the computation of the optimal error predictor requires the solution of two linear programming problems.

The extension of the previous results to multistep predictions is not an easy task when autoregressive components are used. In fact, optimal multistep ahead predictors can be easily computed only if X is defined as a space of deterministic functions. In any case, suboptimal multistep predictors can be obtained by applying repeatedly the optimal one step ahead predictor (12) and substituting the predicted values of the series for the corresponding unknown observations.

3. GMDH predictors

The GMDH is a multilayer self-organizing procedure able to build a polynomial approximation of the relationship between a dependent variable y and a number of independent variables $x_1, \ldots, x_n$. The original idea proposed by Ivakhnenko (1970) has been elaborated and developed by several authors more specifically in the field of system identification and parameter estimation [Duffy and Franklin (1975), Ikeda et al. (1976), Tamura and Kondo (1980), Gallo et al. (1985)]. In modelling and identification contexts, the GMDH appears as a typical black-box approach which may be useful when little a priori knowledge is available about the structure of the data. This may be the case in time series analysis, where frequently scarce information is available on the physical mechanism generating the data. As already mentioned, the GMDH is a self-organizing algorithm which manipulates the independent variables $x_1, \ldots, x_n$ ('inputs') in couples and gives as output a polynomial approximation of the functional relationship

$$y = h(x_1, \ldots, x_n), \tag{13}$$

in the form of a partial discrete Volterra expansion [Gabor et al. (1961)]. Exhibit 1 shows the basic structure of a GMDH algorithm. In Exhibit 1 each block P_2 is a 2nd order complete polynomial in the two corresponding inputs; for example, in the 1st stage we have

$$w_i = P_2(x_j, x_k) = b_{0i} + b_{1i}x_j + b_{2i}x_k + b_{3i}x_j^2 + b_{4i}x_k^2 + b_{5i}x_jx_k,$$

$$i = 1,\dots,m, \quad m = n(n-1)/2, \quad k = 1,\dots,n, \quad j = 1,\dots,k-1. \tag{14}$$

A prerequisite of the basic GMDH is the separation of the available data into two sets (training and testing sets). At each layer, all possible input pairs are generated and the output of each block P_2 is fitted to the training set; the testing set is used to check which of the fitted outputs better explain the data not used in the fitting procedure. In fact, only n out of the $m \geqslant n$ outputs of a stage are chosen as intermediate variables and passed as inputs to the next stage. The procedure can be stopped according to one of the following two criteria. One is based on the rate of decrease of the approximation error so that the algorithm can be stopped when this rate goes below a fixed threshold. The other possible criterion simply takes into account the maximum admissible degree of the polynomial model, so that the algorithm is stopped when the degree of the approximating polynomial reaches that value.

In the above outlined basic structure two major heuristics are present, namely the assumed fixed structure of partial models P_2 (quadratic polynomials) and the undetermined criterion for the separation of data into training and testing sets. With respect to these problems, a revised version of the GMDH has been proposed in Tamura and Kondo (1980), which suggests the use of criteria such as PRESS or AIC to select the structures of partial models at each stage and, at the same time, to use all data both as training and testing sets, so avoiding the problem of separation of the data.

In the context of univariate time series analysis, the GMDH can be used as shown in the following [see also Ikeda et al. (1976), Tamura and Kondo (1980), Gallo et al. (1985) for further details]. The

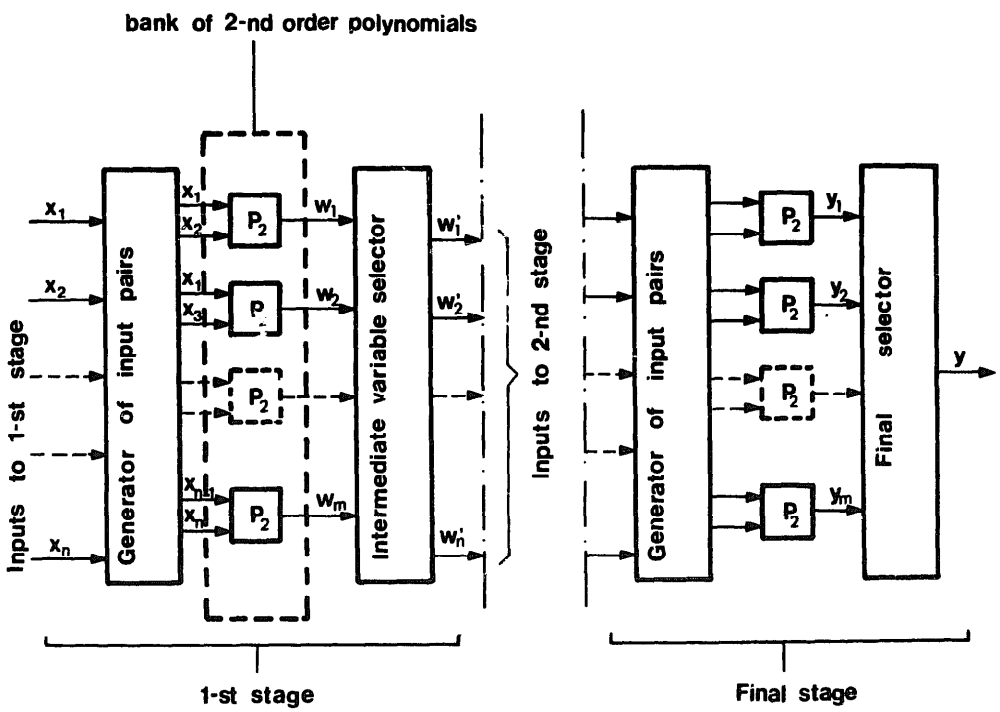

Exhibit 1. Schematic GMDH procedure ($m \geqslant n$).

inputs x_i of the initial stage are assumed as the lagged samples $y(k-i)$ of the dependent variable $y(k)$, while the inputs to successive layers are the lagged samples of some variables selected among the fitted ones of the preceding layers. For each fixed pair of input variables x_i, x_j, i.e., $y(k-i)$, $y(k-i)$, we use the PRESS criterion to select the optimal structure of the corresponding partial model; in particular, the optimal model is chosen as the one which give minimum PRESS in the class of the following four families of models:

$$S_1: y(k) = b_0 + b_1 y(k-i) + b_2 y(k-j), \tag{15}$$

$$S_2: y(k) = b_0 + b_1 y(k-i) + b_2 y^2(k-i), \tag{16}$$

$$S_3: y(k) = b_0 + b_1 y(k-j) + b_2 y^2(k-j), \tag{17}$$

$$S_4: y(k) = b_0 + b_1 y(k-i) + b_2 y(k-j) + b_3 y^2(k-i) + b_4 y^2(k-j) + b_5 y(k-i) y(k-j). \tag{18}$$

The multistage procedure is stopped when all the variables selected as input to the next stage are generated by linear partial models (15); in fact, in such case no further reduction of the PRESS is possible. Once the model structure has been determined the prediction at various lead times is obtained by recursively solving the estimated nonlinear difference equation.

4. Experimental results

Two widely studied time series have been considered to evaluate the proposed methods: the Annual Canadian Lynx Trappings [Campbell and Walker (1977)] and the Wolf Sunspot Numbers [Waldmeier (1961)]. The forecasting performances of some of the best known models available in the extensive literature on the subject have been compared with those obtained by applying an Optimal Error Predictor (OEP) and a GMDH algorithm. In addition, the OEP has been applied to the recently published demographic series of the Australian Births from 1922 to 1976 [McDonald (1981)]. In this case the obtained forecasting results have been compared with those obtained by two univariate statistical models reported in [McDonald (1981)].

The series have been divided in two periods. The first one, which we shall call the fitting period, has been used both to select the model structure and estimate parameters (which in the case of the OEP are the error bound ξ and the weights $g(i)$), while the second one, which we shall call the forecasting period, has been used to test how the selected models perform in forecasting new data. In this last respect, we observe that for the Canadian Lynx case, because almost all of the models available in the literature had been estimated on the whole length of the series, it was necessary for us to reestimate their parameters using only the data included in the fitting period.

Two performance indexes have been used as a measure of the goodness of the forecasts of the models on the forecasting period, namely the prediction mean square error (MSE) and the maximum prediction error in absolute value ($MAXE$) at different lead times. For the first two series we have considered a forecasting horizon equal to an approximate estimate of the cycle of the series (10 years for the Lynx series and 11 years for the Sunspot series); for the third series we have considered the maximum lead time for which forecast probability limits obtained in statistical models are judged useful for demographic analysis, i.e., 4 years [see McDonald (1981)]. To evaluate the behaviour of the different models on the whole forecasting horizon, we have considered a performance index (MSE_{AV}) consisting of the mean square prediction error on the forecasting period averaged on all lead times covering the forecasting horizon.

4.1. Annual Canadian Lynx Trapping (1821–1934)

Six statistical models have been selected among the best known in the literature, four of which are linear and two nonlinear. All of them refer to the logarithmically transformed data and their parameters have been estimated by means of least squares, maximum likelihood or similar techniques.

According to what is done in Gabr and Subba Rao (1981) we have selected the first 100 data of the series as fitting period (1821–1920), while the last 14 data (1921–1934) have been used as forecasting period. Exhibit 2 shows the transformed Lynx series with the fitting and forecasting periods. Among the models reported in the following, M1, M2, M3, M4 and M8 refer to the data after subtraction of the mean value estimated on the fitting period ($\bar{y} = 2.880$), while M5, M6 and M7 refer to the original data shown in Exhibit 2.

As already pointed out before, it must be said that the parameters of models M1, M2, M3, M4 and M5 reported in the references were estimated using all the data points of the series. On the contrary, the parameters reported here have been obtained by only using the fitting period. Nevertheless, several numerical computations have shown that the slightly different values of the parameter estimates, due to the smaller number of data included in the fitting period, do not affect significantly the forecasting performance of the models on the forecasting period.

(*1*) Model M1 [Tong (1977)]
M1 is a linear autoregressive model of order 11.

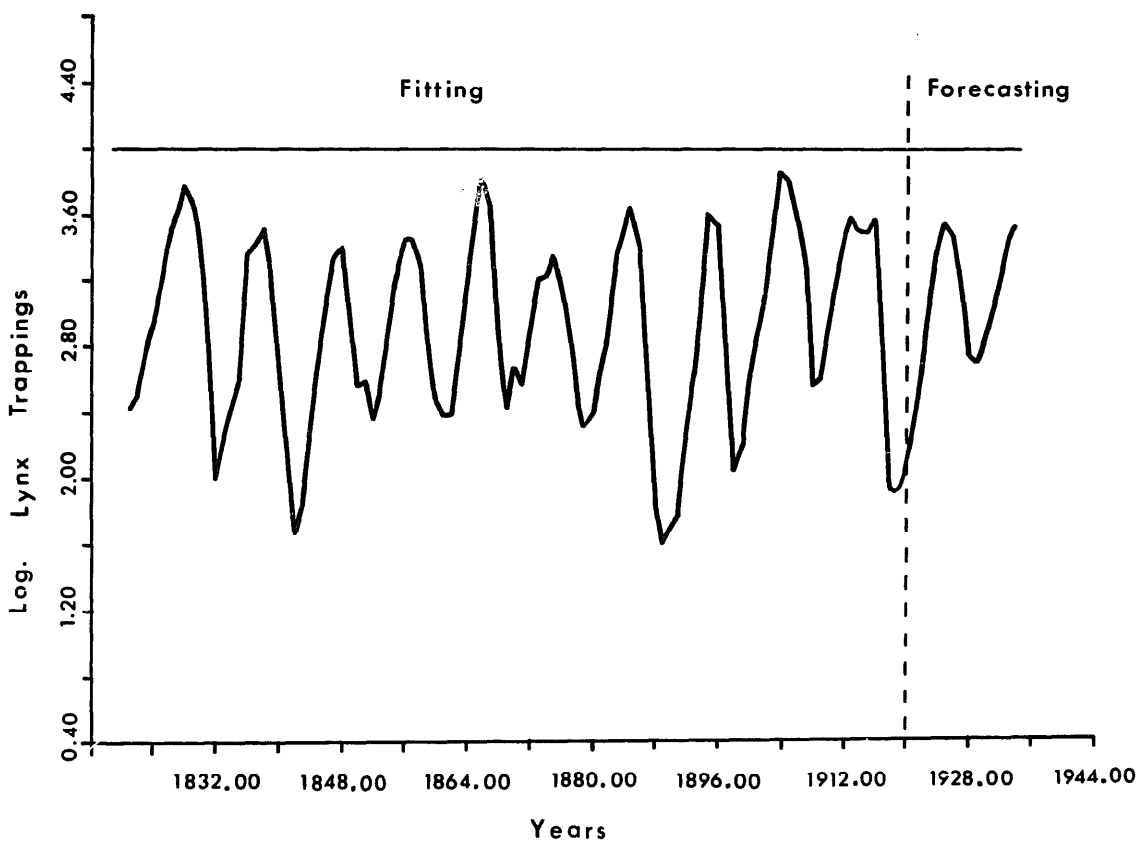

Exhibit 2. Annual Canadian Lynx Trappings (1821–1934).

$$M1: y(k) = +1.128y(k-1) - 0.518y(k-2) + 0.273y(k-3) - 0.350y(k-4) + 0.192y(k-5)$$
$$-0.117y(k-6) + 0.019y(k-7) - 0.030y(k-8) + 0.178y(k-9) + 0.218y(k-10)$$
$$-0.361y-(k-11) + e(k).$$

The obtained residual variance is $\hat{\sigma}_e^2 = 0.0350$.

(2) Model M2 [Campbell and Walker (1977)]
M2 is a linear model consisting of the sum of a harmonic component and an autoregression of order 2.

$$M2: y(k) = 0.998y(k-1) - 0.308y(k-2) - 0.282\cos[(2\pi/9.63)(k-1.32)] + e(k),$$

with residual variance $\hat{\sigma}_e^2 = 0.0415$.

(3) Model M3 [Tong (1977)]
M3 is a linear subset autoregressive model with maximum delay 11.

$$M3: y(k) = 1.149y(k-1) - 0.429y(k-2) + 0.038y(k-4) + 0.405y(k-10)$$
$$-0.410y(k-11) + e(k),$$

with residual variance $\hat{\sigma}_e^2 = 0.0437$.

(4) Model M4 [Priestley (1981)]
M4 belongs to the general class of linear autoregressive-moving average models. The following ARMA(3, 3) has been fitted to the data of the fitting period:

$$M4: y(k) = 1.682y(k-1) - 1.145y(k-2) + 0.1098y(k-3) - 0.479e(k-1)$$
$$-0.160e(k-2) + 0.521e(k-3) + e(k),$$

with residual variance $\hat{\sigma}_e^2 = 0.0456$.

(5) Model M5 [Tong and Lim (1980)]
M5 is a threshold autoregressive model SETAR(2; 8, 3).

$$M5: y(k) = 0.54 + 1.00y(k-1) - 0.167y(k-2) + 0.169y(k-3) - 0.41y(k-4)$$
$$+0.355y(k-5) - 0.32y(k-6) + 0.22y(k-7) + 0.006y(k-8) + e_1(k)$$
$$\text{if} \quad y(k-2) \leqslant 3.1163,$$
$$= 2.55 + 1.46y(k-1) - 1.13y(k-2) - 0.10y(k-3) + e_2(k)$$
$$\text{if} \quad y(k-2) > 3.1163,$$

with residual variance $\hat{\sigma}_{e_{1,2}}^2 = 0.050$.

(6) Model M6 [Gabr and Subba Rao (1981)]
M6 is a subset bilinear model whose parameters have been estimated in Gabr and Subba Rao (1981) on the fitting period.

$$\begin{aligned} M6: y(k) = {} & 1.486292 + 0.77227y(k-1) - 0.091572y(k-2) + 0.083073y(k-3) \\ & -0.261493y(k-4) + 0.225585y(k-9) - 0.245841y(k-12) \\ & -0.7893y(k-3)e(k-9) + 0.4798y(k-9)e(k-9) + 0.3902y(k-6)e(k-2) \\ & +0.1326y(k-1)e(k-1) + 0.07944y(k-2)e(k-7) \\ & -0.3212y(k-4)e(k-2) + e(k), \end{aligned}$$

with residual variance $\hat{\sigma}_e^2 = 0.0223$.

(7) Optimal Error Predictor (OEP)
Several possible structures of the space X have been tested on the fitting period, e.g., deterministic harmonic or polynomial functions, autoregressive functions, mixed deterministic and autoregressive functions. A selection criterion based on one step ahead mean square prediction error (on the fitting period) has allowed us to choose a mixed harmonic–autoregressive structure as the more suitable for representing the data. This means that for each fixed forecast origin m ($100 \leqslant m \leqslant 113$) the following structure of the X space has been considered

$$X = \left\{ x(\cdot): x(k) = a_1 + a_2 \sin \frac{2\pi}{T}k + a_3 \cos \frac{2\pi}{T}k + \sum_{i=1}^{2} b_i \delta(k-i) y(k-i) + \delta(k) e(k), \quad k = 3, \ldots, m+1 \right\}, \tag{19}$$

where T is assumed as the approximate period of the cycle of the series ($T = 10$). Furthermore, the bound ξ on the noise process and the weights $g(i)$ have been assumed so that the uncertainty terms $e(i)$ are the sum of two components

$$|e(i)| = |\xi/g(i)| = \xi_a + \xi_r |y(i)|, \qquad i = 3, \ldots, m+1, \tag{20}$$

where ξ_a and ξ_r are nonnegative real numbers representing the absolute and relative components of the noise. The values of ξ_a and ξ_r obtained on the fitting period of the series are $\xi_a = 0.45$, $\xi_r = 0.0$.

(8) GMDH
The nonlinear structure selected by the procedure described in section 3 on the data of the fitting period after subtraction of the mean value, is

$$\begin{aligned} M8: y(k) = {} & 0.076 + 1.365y(k-1) - 0.772y(k-2) + 0.08y^2(k-1) - 0.427y^2(k-2) \\ & +0.145y(k-1)y(k-2) + e(k), \end{aligned}$$

with residual variance $\hat{\sigma}_e^2 = 0.046$.

Exhibits 3 and 4 show some prediction results obtained by the eight described models. The maximum lead time considered in multistep prediction is 10, i.e., about one cycle. Exhibit 3 reports the prediction mean square errors and maximum errors for lead times 1 and 10. Exhibit 4 reports the average prediction mean square error MSE_{AV} from one to ten steps ahead.

4.2. Yearly Wolf Sunspot Numbers (1770–1892)

The period 1770–1892 has been chosen because the use of these data allows to make a comparison with some statistical models proposed in the literature.

The performances of three among the most known models in the literature have been compared with the results of the OEP and GMDH algorithm. The series has been divided in a fitting period with the first 100 observations and a forecasting period with the last 23 data points (approximately two cycles). Exhibit 5 shows the Sunspot Number series.

(*1*) Model M1 [Box and Jenkins (1970), Granger and Andersen (1978)]
M1 is a linear autoregressive model of second order estimated over the fitting period of the series (1770–1869).

$$M1: y(k) = 14.70 + 1.425y(k-1) - 0.731y(k-2) + e(k),$$

with residual variance $\hat{\sigma}_e^2 = 228$.

(*2*) Model M2 [Granger and Andersen (1978)]
M2 is a subset bilinear model which has been obtained by fitting a bilinear model to the residuals provided by M1 on the fitting period (1770–1869).

$$M2: \begin{cases} y(k) = 14.70 + 1.425y(k-1) - 0.731y(k-2) + e(k), \\ e(k) = -0.0222e(k-2)\eta(k-1) + 0.202e(k-1) + \eta(k). \end{cases}$$

Exhibit 3
Mean square and maximum errors at lead times 1 and 10 for the Lynx series.

Model	$MSE(1)$	$MAXE(1)$	$MSE(10)$	$MAXE(10)$
M1 : AR(11)	0.0208	0.33	0.1750	0.64
M2 : AR(2) + Harmonic	0.0255	0.26	0.1465	0.67
M3 : Subset AR(11)	0.0188	0.26	0.1759	0.60
M4 : ARMA (3, 3)	0.0233	0.31	0.1775	0.69
M5 : SETAR (2; 8, 3)	0.0120	0.21	0.1470	0.79
M6 : Subset Bilinear	0.0148	0.26	0.1714	0.86
M7 : OEP	0.0130	0.24	0.0725	0.46
M8 : GMDH	0.0083	0.22	0.1300	0.86

Exhibit 4
Average mean square errors for predictions from one to ten steps ahead for the Lynx series.

Model	MSE_{AV}
M1	0.1197
M2	0.1232
M3	0.1452
M4	0.1126
M5	0.1004
M6	0.1034
M7	0.0645
M8	0.0649

The residual variance is $\hat{\sigma}_{\eta}^2 = 197$.

(*3*) Model M3 [Tong and Lim (1980)]
M3 is a threshold autoregressive model SETAR(2; 4, 3) fitted on the fitting period (1770–1869).

$$M3: y(k) = 5.2659 + 1.8891y(k-1) - 1.5289y(k-2) + 0.3039y(k-3)$$

$$+0.3387y(k-4) + e_1(k) \quad \text{if} \quad y(k-3) \leqslant 36.6,$$

$$= 0.3900 + 1.1366y(k-1) - 0.3645y(k-2) + 0.0524y(k-3) + e_2(k)$$

$$\text{if} \quad y(k-3) > 36.6,$$

with pooled residual variance $\hat{\sigma}_{e_{1,2}}^2 = 122$.

(*4*) OEP
Also in this case, as for the Canadian Lynx series, a space of mixed harmonic autoregressive functions is found to be the most suitable in modeling the series. The selected structure of the X space is the same as for the Lynx series in (19), with two differences only, i.e., $T = 11$ and $100 \leqslant m \leqslant 122$. The bound ξ and the weights $g(i)$ have been chosen as in the Lynx case (see (20)). The values of ξ_a and ξ_r obtained on the fitting period of the series are $\xi_a = 20.0$, $\xi_r = 0.30$.

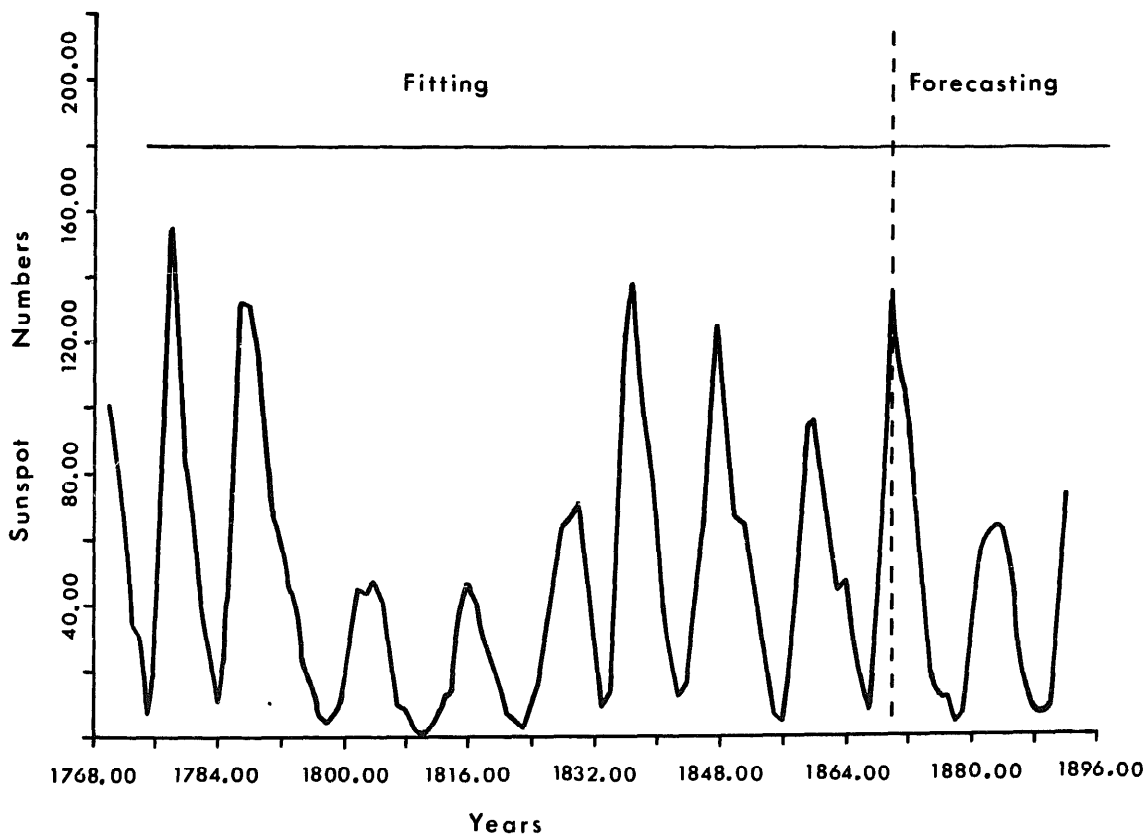

Exhibit 5. Yearly Wolf Sunspot Numbers (1770–1892).

(5) GMDH
The nonlinear structure selected on the data after subtraction of the mean value is

$$M5: y(k) = -3.54 + 1.469y(k-1) - 0.809y(k-2) + 0.0025y^2(k-1) + 0.0125y^2(k-2)$$
$$-0.0148y(k-1)y(k-2) + e(k),$$

with residual variance $\hat{\sigma}_e^2 = 122$.

Exhibits 6 and 7 show the obtained results as in the case of the Lynx series, with the only difference that the maximum prediction lead time is in this case 11.

4.3. Australian Births (1922–1976)

As in McDonald (1981), only the first 44 data points of the series have been used to fit the OEP error bound parameters, while the last 11 data have been considered as forecasting period. Exhibit 8 shows the Australian Births series.

One step ahead forecasting results of the two statistical models reported in McDonald (1981), i.e., a linear ARIMA and a bilinear model, have been compared with those obtained by the OEP.

Multistep prediction performances of the OEP have been compared with those of the linear model ARIMA only.

(1) Model M1 [McDonald (1981)]
M1 is a very simple ARIMA model, i.e., a random walk with drift model.

$$M1: y(k) = 743.0 + y(k-1) + e(k), \qquad \text{with residual variance } \hat{\sigma}_e^2 = 7.118 \cdot 10^6.$$

Exhibit 6
Mean square and maximum errors at lead times 1 and 11 for the Sunspot series.

Model	$MSE(1)$	$MAXE(1)$	$MSE(11)$	$MAXE(11)$
M1: AR(2)	325	47	1062	81
M2: Subset bilinear	276	46	1062	81
M3: SETAR (2; 4, 3)	259	44	1223	76
M4: OEP	219	38	460	47
M5: GMDH	215	42	865	81

Exhibit 7
Average mean square errors for predictions from one to eleven steps ahead for the Sunspot series.

Model	MSE_{AV}
M1	897
M2	887
M3	782
M4	447
M5	728

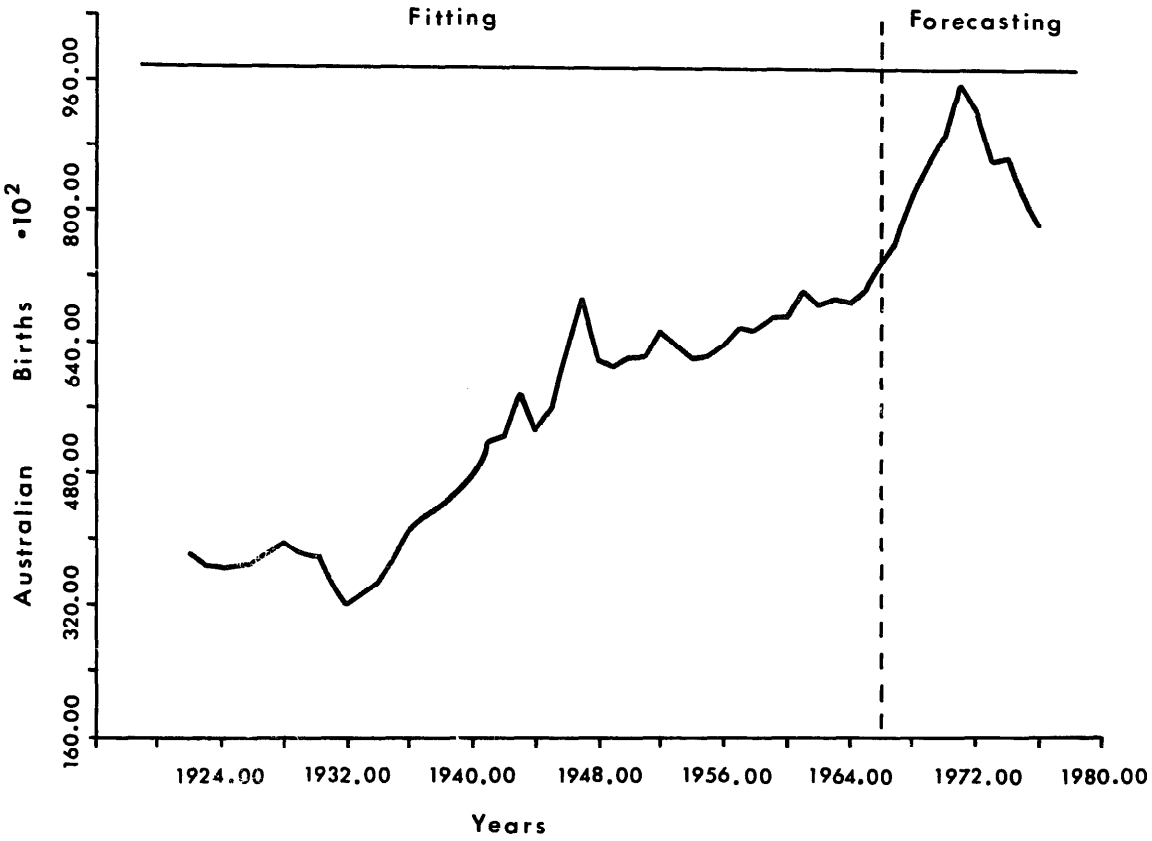

Exhibit 8. Australian Births (1922–1976).

(*2*) Model M2 [McDonald (1981)]
M2 is a bilinear model fitted to the residuals of a random walk with drift model.

$$M2: \begin{cases} y(k) = \alpha_1 + y(k-1) + e(k), \\ e(k) = \alpha_2 e(k-1)\eta(k-2) + \eta(k). \end{cases}$$

The estimated values of the parameters α_1 and α_2 are not given in McDonald (1981), where only one step forecast errors on the forecasting period are reported.

(*3*) OEP
The most suitable structure of the space X selected on the fitting period for $44 \leqslant m \leqslant 54$ is

$$X = \left\{ x(\cdot) : x(k) = a_1 + a_2 k + \sum_{i=1}^{2} b_i \delta(k-i) y(k-i) + \delta(k) e(k), k = 3, \ldots, m+1 \right\}, \tag{21}$$

i.e., a mixed space of linear trend and autoregressive functions.

The bound ξ on the noise process and the weights $g(i)$ have been chosen as in the Lynx case (see (20)). The obtained values of ξ_a and ξ_r are

$$\xi_a = 1000.0, \qquad \xi_r = 0.10.$$

Exhibit 9
Mean square and maximum errors for lead time 1 and 4 for the Australian Births series.

Model	$MSE(1)$	$MAXE(1)$	$MSE(4)$	$MAXE(4)$
M1 : ARIMA (0, 1, 0)	$1.76\ 10^7$	6700	$12.1\ 10^7$	16800
M2 : Bilinear	$1.68\ 10^7$	6300	–	–
M3 : OEP	$1.52\ 10^7$	7100	$8.65\ 10^7$	20600

Exhibit 10
Average mean square errors for predictions from one to four steps ahead for the Australian Births series.

Model	MSE_{AV}
M1	$6.67\ 10^7$
M3	$4.94\ 10^7$

Exhibit 9 shows the obtained maximum and mean square prediction errors at lead times 1 and 4, while Exhibit 10 reports the average mean square error from 1 to 4 steps ahead.

5. Concluding remarks

The results presented in the preceding section clearly show that GMDH and OEP obtain one step ahead forecasts of at least the same degree of accuracy as those of some well known statistical models. In addition, with regard to multistep forecasts, the OEP provides significantly better results than other methods examined. This result is not too surprising because the OEP solves the forecasting problem avoiding the preliminary step of model parameter estimation. In fact, it is well known that the wider the forecasting horizon, the stronger the interaction between model fitting and prediction with the estimated model. In contrast, both the examined statistical models and the GMDH do not show such features of robustnesses to changes of prediction lead times or of data characteristics. Nonlinear statistical models, which give good results in one-step ahead forecasting, result in inaccurate multistep prediction; for example, the subset bilinear model, which gives good results in one-step ahead prediction on the Lynx series, is ranked fourth for the Sunspot series and provides unsatisfactory multistep predictions. The GMDH algorithm obtains excellent results for one step ahead forecasts but gives less accurate multistep forecasts on the Sunspot series.

Exhibits 4, 7 and 10 provide evidence that, among statistical models, simple linear models reveal a remarkable predictive capability with respect to that of much more complicated (non linear) models; this fact substantiates the concept that non-parsimonious parameterizations are not suitable especially for multistep forecasting.

It could be argued that statistical adaptive algorithms may be advantageously used, especially in the case of Sunspot Numbers, due to the nonstationarities of the process. However, though recursive estimation schemes have been widely studied and developed, the relative shortness of the examined series prevents their application, due to problems of speed of convergence. The OEP does not suffer from this drawback as it estimates adaptively future values of the series, keeping fixed only the noise process bounds. Also several additional trials performed on the series have shown a low sensitivity of the prediction results to variations of the error bounds ξ_a and ξ_r obtained in the fitting stage; this again indicates good robustness properties of the method.

From a computational point of view, the OEP is revealed as a time consuming algorithm due to the need to solve two linear programming problems at each time period for each forecast origin and for each prediction lead time. On the contrary, stochastic difference equations are particularly well

suited for predictions, which can be easily obtained by recursively solving the estimated difference equation.

A further appealing feature of the optimal algorithm approach consists in its robustness to modelling errors, non linearities or parameter variations. In fact, unlike statistical approaches, all of these uncertainties can be handled by suitable tuning of only the noise bounds without dropping the optimality of the results.

The above considerations have basically motivated the innovative approach of the Optimal Error Predictor. The theoretical expectations on the performance of the two presented estimators (especially the OEP) appear to be confirmed by the first numerical examples presented in this paper. These preliminary results encourage further work in the direction of a wider comparison of the forecasting performance of these methods with those of classical statistical techniques.

References

Akaike, H., 1969, Fitting autoregressive models for prediction, Ann. Inst. Stat. Math. Tokyo 21, 243.

Allen, D.M., 1974, The relationship between variable selection and data augmentation and a method for prediction, Technometrics 16, 125–127.

Belforte, G., M. Milanese and R. Tempo, 1982, Optimal algorithms theory and estimation with unknown but bounded errors, Proceedings of the 21st IEEE Conference on Decision and Control, Orlando, FL.

Box, G.E.P. and G.M. Jenkins, 1976, Time series analysis: Forecasting and control, revised ed. (Holden Day, San Francisco, CA).

Campbell, M.J. and A.M. Walker, 1977, A survey of statistical work on the Mackenzie river series of annual Canadian Lynx trappings for the years 1821–1934 and a new analysis, Journal of the Royal Statistical Society, Series A 140, 411–431.

Duffy, J.J. and M.A. Franklin, 1975, A learning identification algorithm and its application to environmental system, IEEE Transactions on Systems, Man and Cybernetics SMC-5, 226–240.

Gabor, D., W.P.L. Wilby and R. Woodcock, 1961, A universal nonlinear filter predictor and simulator which optimizes itself by a learning process, IEE Proceedings 108 B, 422–438.

Gabr, M.M. and T. Subba Rao, 1981, The estimation and prediction of subset bilinear time series models with applications, Journal of Time Series Analysis 2, 155–171.

Gallo, P., R. Genesio and A. Vicino, 1985, Nonlinear structure identification for multistep prediction via a modified GMDH algorithm, Proceedings of VII IFAC Symposium on Identification and System Parameter Estimation, York.

Granger, C.W.J. and A.P. Andersen, 1978, Introduction to bilinear time series models (Vandenhoeck and Ruprecht, Gottingen).

Ikeda, S., M. Ochiai and Y. Sawaragi, 1976, Sequential GMDH algorithm and its application to river flow prediction, IEEE Transactions on Systems, Man and Cybernetics SMC-6, 473–479.

Ivakhnenko, A.G., 1970, Heuristic self-organization in problems of engineering cybernetics, Automatica 6, 207–219.

Kashyap, R.L. and A.R. Rao, 1976, Dynamic stochastic models from empirical data (Academic Press, New York).

McDonald, J., 1981, Modeling demographic relationships: An analysis of forecast functions for Australian births, Journal of the American Statistical Association 76, 782–792.

Micchelli, C.A. and T.J. Rivlin, 1977, A survey of optimal recovery, in: Micchelli and Rivlin, eds., Optimal estimation in approximation theory (Plenum Press, New York).

Milanese, M. and R. Tempo, 1985, Optimal algorithms theory for robust estimation and prediction, IEEE Transactions on Automatic Control AC-30, 730–738.

Milanese, M., R. Tempo and A. Vicino, 1984, Robust time series prediction by optimal algorithms theory, Proceedings of the 23rd IEEE Conference on Decision and Control, Las Vegas, NV.

Parzen, E., 1974, Some recent advances in time series modeling, IEEE Transactions on Automatic Control AC-19, 723–730.

Priestley, M.B., 1981, Spectral analysis and time series, Vol. I (Academic Press, New York).

Tamura, H. and T. Kondo, 1980. Heuristic free group method of data handling algorithm of generating optimal partial polynomials with application to air pollution prediction, International Journal of Systems Science 11, 1095–1111.

Tong, H., 1977, Some comments on the Canadian Lynx data, Journal of the Royal Statistical Society, Series A, 140, 432–436.

Tong, H. and K.S. Lim, 1980, Threshold autoregression, limit cycles and cyclical data, Journal of the Royal Statistical Society, Series B 42, 245–292.

Traub, J.F. and H. Wozniakowski, 1980, A general theory of optimal algorithms (Academic Press, New York).

Waldmeier, M., 1961, The sunspot activity in the years 1610–1960 (Schulthess, Zurich).

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